■ PROBABILITY & STATISTICS COURSE PROJECT

AudioMental

Statistical & Inferential Modeling of Music Genre Preferences on Psychological Well-Being (Anxiety, Depression, Insomnia & OCD)

Project Research Authors

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Course Context & Dataset Specifications

Course:	Probability & Statistics (Class Review & Evaluation)
Dataset:	Kaggle Music & Mental Health Survey (MxMH)
Sample Size:	N = 735 Validated Survey Respondents
Variables:	33 Evaluated Features (Mental Health Scales 0-10, Habits, Genres)
Core Methods:	Continuous Gaussian PDF, Bivariate OLS, Welch's t-Test, Chi-Square, Binomial
Deliverables:	Live Interactive Dashboard + Offline Standalone + Slide Deck + PDF

EXECUTIVE SUMMARY

Research Motivation & Scientific Objectives

Investigating whether subjective musical taste has statistically measurable links with clinical mental health scores.

1. Problem Formulation

RQ1 & RQ2

Primary Research Questions:

- Does daily music listening duration correlate linearly with self-reported anxiety severity?
- Do workplace listeners experience lower psychological strain than non-workplace listeners?
- Does preference for intense genres (e.g., Metal, Rock) associate with higher incidence of clinical anxiety or depression?

Scientific Importance:

Music therapy is widely cited, but quantitative probabilistic foundations across large-sample real-world surveys remain under-analyzed.

2. Statistical Framework

Curriculum

Curriculum Topic Alignments:

1. Exploratory Location & Dispersion: Sample mean, variance, IQR, Tukey outlier fences.
2. Continuous Modeling: Normal distribution fitting and empirical density overlays.
3. Bivariate Association: Pearson product-moment vs. Spearman rank correlation.
4. Parametric Modeling: Ordinary Least Squares (OLS) bivariate regression.
5. Two-Sample Testing: Welch unequal-variance t-test with Satterthwaite df.
6. Contingency Analysis: Pearson Chi-Square (χ^2) test of independence.
7. Discrete Modeling: Binomial PMF & Bayes conditional probability theorem.

3. Summary of Findings

Empirical

Key Empirical Takeaways:

- Weak Linear Trend: Pearson $r = 0.0497$ ($p = 0.1787$). Listening hours do not linearly dictate anxiety.
- Monotonic Rank Trend: Spearman $\rho = 0.0941$ ($p = 0.0107$*). Indicates subtle monotonic rank association.
- Workplace Invariance: Welch $t = -0.9791$ ($p = 0.3285$). Listening while working does not significantly alter anxiety.
- Genre Independence: $\chi^2 = 6.9182$ ($df = 6$, $p = 0.3285$). Genre preference is independent of high anxiety.
- Positive Solace: 73.7% survey-wide report music improves their psychological state.

■ SCIENTIFIC TAKEAWAY FOR COURSE REVIEW:

Self-reported emotional solace does NOT translate into disparate baseline mental health severities. Statistical tests confirm that listeners of all genres share comparable psychological distributions.

DATA PIPELINE

Dataset Anatomy & Robust Preprocessing Pipeline

Transformation of raw survey records into structured, distribution-ready statistical vectors.

Survey Data Cleaning & Filtering Protocol

ETL Pipeline

Raw Ingestion: 736 survey submissions collected across diverse international demographics.

Validation & Dropping: Removed row with missing primary genre / unparseable numeric bounds (Final N = 735).

Continuous Bounds: Daily listening hours truncated and validated against physical day boundaries (0 - 24 hrs).

Discrete Likert Scales: Anxiety, Depression, Insomnia, OCD scored integer discrete [0, 10].

Nominal Categorical: Primary Genre (16 discrete classes: Rock, Pop, Metal, Classical, etc.).

Binary Dichotomy: Listen while working? [Yes (n=578, 78.6%), No (n=154, 21.0%), Unknown (n=3)].

Perceived Solace: Music Effects [Improve (n=542, 73.7%), No Effect (n=169, 23.0%), Worsen (n=17, 2.3%)].

Format Parity: Simultaneously serialized into JSON for web runtime and CSV for base R statistical verification.

Global Empirical Summary Metrics (N = 735)

Parameters

Metric Variable	Sample Mean (x■)	Sample SD (s)	Median	IQR [Q1, Q3]
Anxiety Score (0-10)	5.83	2.79	6.0	[4.0, 8.0] (4.0)
Depression Score (0-10)	4.80	3.03	5.0	[2.0, 7.0] (5.0)
Insomnia Score (0-10)	3.74	3.09	3.0	[1.0, 6.0] (5.0)
OCD Score (0-10)	2.63	2.85	2.0	[0.0, 4.0] (4.0)
Daily Listening Hours	3.71 hrs	3.03	3.0 hrs	[2.0, 5.0] (3.0)
Participant Age	25.2 yrs	12.05	21.0 yrs	[18.0, 28.0] (10.0)

SAMPLE STANDARD DEVIATION (BESSEL-CORRECTED)

```
s = sqrt( (1 / (n - 1)) * sum( (x_i - mean(x))^2 ) )
```

DESCRIPTIVE STATISTICS

Location, Dispersion & Five-Number Summary Analysis

Evaluating central tendency and non-parametric spread across psychological dimensions.

Theoretical Formulas & Foundations

Theory

SAMPLE MEAN (FIRST CENTRAL MOMENT)

```
x_bar = (1 / n) * sum(x_i, i=1..n)
```

SAMPLE VARIANCE (BESSEL'S CORRECTION)

```
s^2 = (1 / (n - 1)) * sum( (x_i - x_bar)^2 )
```

TUKEY'S INTERQUARTILE RANGE & OUTLIER FENCES

```
IQR = Q3 - Q1   |   Whiskers: [Q1 - 1.5*IQR, Q3 + 1.5*IQR]
```

- Robustness Considerations: Means and variances are susceptible to extreme values (e.g. participant reporting 24h listening).
- Median & IQR Resilience: The median (Q2) remains resilient to skewed tails, making it ideal for discrete 0-10 Likert scales.
- Outlier Classification: Any observation outside [Q1 - 1.5 IQR, Q3 + 1.5 IQR] is isolated as an outlier in our interactive boxplots.
- Implementation in AudioMental: Live boxplots feature an interactive Jitter toggle that renders each participant dot overlaid on top of quartiles.

Two-Sample Boxplot Quartiles: Work vs Non-Work

Application

Comparison of Anxiety Scores grouped by 'Listens while working':

Statistic	Listens While Working (YES)	Does NOT Listen (NO)
Sample Size (n)	578 participants (78.6%)	154 participants (21.0%)
Sample Mean (x■)	5.88	5.62
Sample Std Dev (s)	2.76	2.90
Minimum (Min)	0.0	0.0
First Quartile (Q1)	4.0	3.0
Median (Q2)	6.0	6.0 (Identical)
Third Quartile (Q3)	8.0	8.0 (Identical)
Maximum (Max)	10.0	10.0
Interquartile Range	IQR = 4.0	IQR = 5.0

Statistical Finding:

Both cohorts exhibit identical median scores of 6.0 and matching Q3 boundaries. Workplace listening neither elevates nor depresses clinical anxiety distributions.

Continuous Probability Density Modeling (Gaussian Fit)

Superimposing theoretical Normal distributions N(μ, σ²) onto discrete empirical histograms.

Theoretical Normal Density & Scaling

Mathematical Model

GAUSSIAN PROBABILITY DENSITY FUNCTION

```
f(x; mu, sigma) = (1 / (sigma * sqrt(2*pi))) * exp( -((x - mu)^2) / (2*sigma^2) )
```

DISCRETE HISTOGRAM FREQUENCY SCALING EQUATION

```
h(x) = N * Delta_x * f(x; mu, sigma)
```

- Dimensionality: Anxiety, Depression, Insomnia, and OCD are recorded on 0-10 discrete scales, while daily listening is a continuous non-negative real.
- Binning Protocol: In AudioMental's interactive Studio, users can vary the bin count k in [5, 20]. The step width Delta_x = (x_max - x_min) / k.
- Scaled Density Curve: Multiplying the PDF f(x) by N * Delta_x aligns the continuous curve area directly with discrete bar count frequencies.
- Goodness of Visual Fit: Anxiety exhibits a slight negative skew with a mode at 6-7, while OCD shows a strong right skew (peak at 0-1).

Interactive Histogram Studio Capabilities

Live Feature

Parametric Bin Slider:
Dynamically re-aggregates 735 observations from 5 to 20 bins in real time.

Normal Curve Toggle:
Computes live sample mean and standard deviation, overlaying glowing SVG Gaussian curves.

Mean & Median Lines:
Draws vertical reference lines: cyan dashed line for sample mean, amber for sample median.

Hover Crosshair Inspector:
Hovering over any bin displays exact interval [a, b), absolute frequency, and percentage share.

Full Dimension Switching:
Seamlessly toggles between Anxiety, Depression, Insomnia, OCD, and Listening Hours.

■ Crisp Vector SVG Export Engine:

Users can click 'Export SVG' to download mathematically precise vector graphics directly into research reports and course slide decks without pixelation.

BIVARIATE ANALYSIS

Correlation Analysis: Pearson Linear vs. Spearman Rank

Evaluating monotonic vs linear dependency between daily music listening hours and anxiety.

Pearson Product-Moment Correlation (r)

Parametric

PEARSON CORRELATION COEFFICIENT

```
r = Cov(X,Y) / (s_X * s_Y) = sum((x_i - x_bar)(y_i - y_bar)) / sqrt(sum(dx^2)*sum(dy^2))
```

Empirical Finding: r = +0.0497 (t = 1.35, p = 0.1787)

- Hypothesis: H0: rho = 0 (No linear correlation) vs H1: rho != 0
- Critical Value: For df = 733 and alpha = 0.05, critical t = 1.963
- Decision: t = 1.35 < 1.963, p-value = 0.1787 > 0.05
- Statistical Verdict: Fail to Reject H0. There is no statistically significant linear relationship between hours spent listening to music and anxiety.
- Coefficient of Determination: R^2 = r^2 = 0.0025. Only 0.25% of the total variance in anxiety scores can be explained by daily listening duration.

Spearman Rank-Order Correlation (rho)

Non-Parametric

SPEARMAN MONOTONIC RANK-ORDER FORMULATION

```
rho = 1 - ( 6 * sum(d_i^2) ) / ( n * (n^2 - 1) )
```

Empirical Finding: rho = +0.0941 (p = 0.0107 *)

- Hypothesis: H0: No monotonic association between ranks.
- Decision: p-value = 0.0107 < alpha = 0.05. Reject H0.
- Statistical Verdict: Significant positive monotonic trend detected.
- **Key Academic Rigor - Why does Spearman detect significance while Pearson fails?**

1. Robust to Ties & Non-Normality: The 0-10 anxiety scale has heavy discrete ties.
2. Non-Linear Monotonicity: Higher listening hours weakly accompany higher rank orders of anxiety, without obeying a strict straight-line proportional increment.

REGRESSION MODELING

Ordinary Least Squares (OLS) Bivariate Regression

Quantifying the linear predictor model between listening duration and psychological distress.

Mathematical Formulation & Parameter Estimation

OLS Derivation

SIMPLE LINEAR REGRESSION MODEL

```
y_hat = beta_0 + beta_1 * x
```

OLS NORMAL EQUATION PARAMETER ESTIMATORS

```
beta_1 = Cov(X,Y) / Var(X)   |   beta_0 = y_bar - beta_1 * x_bar
```

Slope (beta_1): beta_1 = 0.0458. For every additional hour of music listened daily, predicted anxiety rises by just 0.046 points.

Intercept (beta_0): beta_0 = 5.6698. The baseline predicted anxiety score for an individual listening to zero music hours.

Fitted Model: Anxiety_Hat = 5.6698 + 0.0458 * (Hours)

ANOVA F-Statistic: F(1, 733) = 1.815, p-value = 0.1787. Overall model regression is not statistically significant.

Residual Variance: Residual Standard Error: s_e = 2.788 on 733 degrees of freedom.

Interactive Scatter Studio & Participant Inspector

Application

N=735 Circle Rendering:
Renders all 735 surveyed participants as interactive vector circles with customizable opacity & size.

Per-Participant Inspection:
Hovering over any dot displays their Participant ID, Age, Daily Hours, Genre, Workplace status, and Perceived Effect.

Real-Time Genre Subsetting:
Filter by Rock, Pop, Metal, Classical, etc. Live OLS recalculates slope, intercept, and r instantly.

Regression Line Overlay:
Renders the glowing OLS linear equation badge with interactive toggles for visual assessment.

■■ Practical Interpretation for Course Defense:

While pop culture often assumes excessive music consumption reflects depressive states, OLS modeling proves that daily listening duration explains less than 0.3% of anxiety. Individual psychological state is dictated by factors far beyond daily listening hours.

Two-Sample Welch's t-Test & Confidence Intervals

Testing whether listening while working causes statistically significant shifts in anxiety.

Hypothesis Formulation & Welch-Satterthwaite

Formulation

TWO-TAILED HYPOTHESIS FORMULATION

```
H0: mu_work = mu_nowork   vs.   H1: mu_work != mu_nowork
```

WELCH'S TWO-SAMPLE T-STATISTIC

```
t = (x_bar1 - x_bar2) / sqrt( s1^2 / n1 + s2^2 / n2 )
```

WELCH-SATTERTHWAITE EFFECTIVE DEGREES OF FREEDOM

```
df = (s1^2/n1 + s2^2/n2)^2 / [ (s1^2/n1)^2/(n1-1) + (s2^2/n2)^2/(n2-1) ]
```

- Group 1 (Works to Music): n1 = 578, x_bar1 = 5.88, s1 = 2.76
- Group 2 (No Music at Work): n2 = 154, x_bar2 = 5.62, s2 = 2.90
- Welch's Degrees of Freedom: df = 230.6 (Robust to unequal variances)
- Test Statistic Result: t = -0.9791

Decision Boundaries & 95% Confidence Intervals

Decision

Significance Threshold:
alpha = 0.05 (Two-Tailed Rejection Boundary: alpha/2 = 0.025 per tail)

Critical Values:
t_critical = +/- 1.970

Observed Test Stat:
t = -0.9791 falls comfortably inside acceptance region [-1.970, +1.970].

p-Value:
p = 0.3285 > 0.05. High probability of observing this data under H0.

Statistical Decision:
FAIL TO REJECT NULL HYPOTHESIS (H0).

Cohen's d Effect Size:
d = (x_bar1 - x_bar2) / s_pooled = 0.09 (Negligible effect size < 0.20).

95% POPULATION DIFFERENCE CONFIDENCE INTERVAL

```
95% CI for (mu1 - mu2) = [-0.772, +0.260]
```

CATEGORICAL ANALYSIS

Contingency Tables & Chi-Square (χ^2) Independence Test

Evaluating whether primary music genre preference associates with high clinical anxiety severity.

Chi-Square Test Formulation & Contingency Table

Independence

NULL HYPOTHESIS STATEMENT

H0: Genre Preference and Anxiety Severity are Independent

PEARSON'S CHI-SQUARE TEST STATISTIC

chi^2 = sum_i sum_j ((O_ij - E_ij)^2 / E_ij)

EXPECTED CELL FREQUENCIES UNDER INDEPENDENCE

E_ij = (Row_i * Col_j) / N

- Contingency Matrix: 7 Genres (Classical, EDM, Metal, Pop, Rock, Video Game, Other) x 2 Anxiety Tiers (High >= 6 vs Normal < 6).
- Total Sample N = 735 respondents across contingency cells.
- Assumption Verification: All expected cell frequencies E_ij > 5, satisfying Cochran's rule for valid Chi-square asymptotic approximation.

Test Results & Cramer's V Effect Size

Findings

Chi-Square Statistic:
X^2 = 6.9182

Degrees of Freedom:
df = (r - 1) * (c - 1) = (7 - 1) * (2 - 1) = 6

p-Value:
p = 0.3285 > 0.05

Statistical Verdict:
FAIL TO REJECT NULL HYPOTHESIS (H0). Music genre and high anxiety classification are statistically independent.

Cramer's V Effect Size:
V = sqrt(chi^2 / (n * min(r-1, c-1))) = sqrt(6.918 / 735) = 0.097

Magnitude:
Cramer's V = 0.097 represents a negligible association (< 0.10).

Academic Meaning for Teacher Review

Debunks the common cultural assumption that listeners of heavier genres (e.g., Metal, Rock) possess disproportionately higher clinical anxiety rates.

Discrete Probability: Binomial Distribution PMF & CDF

Modeling random variable X ~ Bin(n, p) for clinical risk forecasting in survey cohorts.

Binomial Model Formulation & Moments

PMF & Moments

BINOMIAL PROBABILITY MASS FUNCTION (PMF)

```
P(X = k) = C(n, k) * p^k * (1 - p)^(n - k)
```

EXPECTED VALUE, VARIANCE & STANDARD DEVIATION

```
E[X] = n * p   |   Var(X) = n * p * (1 - p)   |   sigma = sqrt(Var(X))
```

- **Random Variable Definition: X = Number of high-anxiety respondents (score >= 6) in a random sample of size n.**
- **Empirical Parameter Estimation: Base sample proportion p_hat = 443 / 735 = 0.6027 (60.27%).**
- Baseline Scenario (n = 10 trials):
- Expected Value: E[X] = 10 * 0.6027 = 6.03 respondents
- Variance: Var(X) = 10 * 0.6027 * (1 - 0.6027) = 2.395
- Standard Deviation: sigma = sqrt(2.395) = 1.548
- Modal Probability: Peak probability occurs at k = 6 with P(X = 6) = 0.2508 (25.08%).
- Extreme Event Improbability: P(X = 0) = (1 - 0.6027)^10 = 0.000088 (< 0.01%).

Interactive PMF Calculator & Cumulative Bounds

Live Feature

Dynamic Trials Slider (n in [5, 30]):
Allows users to simulate classroom sample sizes from n=5 to n=30 dynamically.

Success Probability Slider (p):
Permits adjusting base rate from 5% to 95% to model custom sub-populations.

Target Threshold Slider (k):
Highlights target bin k in neon purple while computing both PMF P(X=k) and CDF P(X <= k).

Live Normal Approximation:
For large n * p >= 10, the Binomial converges towards Gaussian distribution via the Central Limit Theorem (CLT).

CUMULATIVE PROBABILITY FORMULATION

```
Cumulative CDF: P(X <= 6) = sum( C(10,i) * p^i * (1-p)^(10-i), i=0..6 ) = 60.9%
```

Bayesian Probability & Conditional Likelihood Modeling

Updating prior probabilities to evaluate genre-conditional anxiety risk: P(High Anxiety | Genre).

Bayes' Law & Prior-to-Posterior Updating

Bayes Formulation

BAYES' THEOREM FORMULATION

$$P(A \mid B) = [\, P(B \mid A) * P(A) \,] / P(B)$$

LAW OF TOTAL PROBABILITY

$$P(B) = P(B \mid A)*P(A) + P(B \mid A^c)*P(A^c)$$

- Event A: Participant reports High Anxiety (Score >= 6). Global Prior P(A) = 0.6027.
- Event B: Participant's favorite music genre is Genre G.
- Posterior P(A | B): The updated conditional probability that a listener reports high anxiety given their primary genre.
- Bayes Factor / Relative Risk: Ratio of posterior to prior: RR = P(A | B) / P(A).

Ranked Conditional Probabilities Across Genres

Ranked Likelihood

Genre (Class)	n	P(High Anxiety | Genre)	Relative Risk (vs 60.3%)
Folk	30	76.7%	+27.3% (Highest Risk)
Lofi	10	70.0%	+16.1%
Pop	114	66.7%	+10.7%
Rock	188	64.4%	+6.8%
Metal	88	61.4%	+1.9%
Survey Baseline	**735**	**60.3%**	**1.00 (Prior Baseline)**
Classical	53	56.6%	-6.1%
Video Game Music	44	54.5%	-9.6%
Latin	3	33.3%	-44.8% (Lowest Sample)

Important Bayes Reasoning for Teacher Defense:

While Folk has the highest conditional rate (76.7%), its small sample size (n=30) means Rock (n=188, 64.4%) contributes far more absolute high-anxiety listeners.

SOFTWARE ENGINEERING

Interactive Mathematical Visualizer & UI Architecture

How AudioMental translates complex statistical theory into responsive, GPU-accelerated visualizations.

Fluid 60fps Shape Morphing & Animation Math

Interpolation

CUBIC EASE-OUT TWEENING FUNCTION

```
ease(t) = 1 - (1 - t)^3   |   x(t) = x_start + (x_target - x_start) * ease(t)
```

POLAR-TO-CARTESIAN RADAR PROJECTION MATH

```
x_i = cx + r_i * cos(theta_i)   |   y_i = cy - r_i * sin(theta_i)
```

Continuous Polar Morphing:
When switching between genres (e.g. Gospel to Rock), the 4-axis radar polygon vertices do not teleport—they fluidly interpolate coordinates frame-by-frame over 480ms.

Donut Arc Trigonometry:
Circumference C = 2*pi*r. Stroke dash-arrays and dash-offsets glide continuously across mental health improvement tiers.

Rolling Odometer Numbers:
KPI values (means, sample size, hours) smoothly roll up from 0 using requestAnimationFrame eased counters.

System Architecture & Offline Reliability

Architecture

Zero External Dependencies:
Built entirely with native HTML5, Vanilla JavaScript, and SVG. Requires zero CDN script downloads, operating reliably in air-gapped sandboxes.

Dual Distribution Models:
Serves both a live HTTP server on port 8080 and a fully self-contained offline artifact HTML file with inlined data.

Silky Tab-to-Tab Transitions:
Features crossfading .view-panel states that glide out (-8px) and glide in (+14px) smoothly when navigating between 6 primary view tabs.

Slow Staggered Leaderboards:
Leaderboard bars expand from 0% with staggered delays (0.85s ease with index * 45ms offset), creating an aesthetic race-to-rank visual.

Interactive Graph Studio Modal:
Clicking any graph across the application opens a 2-column studio with live parameter controls (bins, alpha, OLS toggles, sample sizes, and vector SVG exports).

SUMMARY OF FINDINGS

Comprehensive Statistical Hypothesis Testing Matrix

Consolidated empirical verification across all primary research questions and statistical tests.

Probability & Statistics Hypothesis Verification Matrix

Summary

Statistical Method	Null Hypothesis (H0)	Test Statistic	p-Value	Decision (alpha=0.05)	Academic Conclusion
Welch's Two-Sample t-Test	mu_work = mu_nowork	t = -0.9791 (df=230.6)	0.3285	**Fail to Reject H0**	Workplace listening does not alter anxiety.
Chi-Square Independence	Genre & Anxiety Independent	chi^2 = 6.9182 (df=6)	0.3285	**Fail to Reject H0**	Music genre preference is independent of anxiety.
Pearson Correlation	rho = 0 (Linear correlation)	r = +0.0497 (t=1.35)	0.1787	**Fail to Reject H0**	Listening hours do not linearly dictate anxiety.
Spearman Rank Correlation	rho_s = 0 (Monotonic trend)	rho = +0.0941	0.0107 *	**Reject H0**	Weak positive monotonic rank association exists.
Binomial PMF Model	p = 0.50 (Chance majority)	z = 5.57 (p_hat=0.603)	< 0.0001 ***	**Reject H0**	Statistically significant majority report anxiety >= 6.
One-Way ANOVA	mu_1 = mu_2 = ... = mu_k	F = 1.34 (df=15, 719)	0.1712	**Fail to Reject H0**	Mean anxiety does not differ significantly across genres.

■ Core Conclusion for Probability & Statistics Review:

Our statistical investigation demonstrates that subjective musical preferences act as a therapeutic coping mechanism (73.7% reported improvement), but baseline anxiety and depression distributions remain statistically invariant across genres. Music is a shared universal outlet rather than a clinical differentiator.

COURSE SUBMISSION

Project Authors, Deliverables & Repository Access

Final submission details for Probability & Statistics Course Evaluation.

Project Authors & Student Identification

Authors

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Course Context:

Subject: Probability & Statistics (Course Evaluation)

Project: AudioMental (MxMH Dataset Empirical Analysis)

Project Repository & Inspection Deliverables

Deliverables

Interactive Web Dashboard:
`http://localhost:8080 (Served via Python HTTP Server)`

Standalone Offline Dashboard:
`genre_stats_dashboard.html (Zero external dependencies)`

Course Presentation Deck:
`presentation.html (Interactive HTML slide deck with Print-to-PDF)`

Course Presentation PDF:
`AudioMental_ProbStats_Presentation.pdf (Widescreen 16:9 PDF)`

Base Dataset:
`Data/mxmh_survey_results.csv (N = 735 cleaned entries)`

Verification Scripts:
`additional_analysis.R & web_dashboard/prepare_data.py`

Thank You!

Questions & Discussion Open for Course Evaluation